Eviatar Bach (she/her)

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eviatarbach.com

Mathematics of inference and learning for dynamical systems: data assimilation, Bayesian inverse problems, ensemble methods, and probabilistic forecasting, with applications to weather and climate prediction.

Professional Appointments

2024– **Lecturer in Mathematics of Environmental Data Science**
UKRI Future Leaders Fellow
Department of Meteorology & Department of Mathematics and Statistics
University of Reading, Reading, UK

Steering Committee of the Centre for Mathematics of Planet Earth (CMPE).

Member of the Data Assimilation Research Centre (DARC) and the National Centre for Earth Observation (NCEO).

Outreach Officer for the Centre for Doctoral Training in Mathematics for Our Future Climate.

Fellow of the Higher Education Academy (FHEA).

2022–4 **Postdoctoral Research Fellow**
Climate Modeling Alliance (CliMA), Department of Environmental Science and Engineering, & Department of Computing and Mathematical Sciences
California Institute of Technology (Caltech), Pasadena, USA

Stanback Postdoctoral Fellow in Global Environmental Science under the supervision of Tapio Schneider and Andrew Stuart.

2021–2 **Postdoctoral Research Fellow**
Laboratoire de Météorologie Dynamique
École Normale Supérieure, Paris, France

Make Our Planet Great Again Postdoctoral Fellow under the supervision of Michael Ghil.

Education

2017–21 **PhD in Atmospheric and Oceanic Science**
University of Maryland, College Park, College Park, USA
Supervisors: Eugenia Kalnay (primary) and Safa Mote (secondary).

2012–7 **BSc in Physics and Computer Science (Honours), Minor in Mathematics**
University of British Columbia, Vancouver, Canada

Grants

2026–30/33 **UK Research and Innovation (UKRI)** (£1.66m fEC)
Future Leaders Fellowship, Fellow (sole PI), "Machine learning for next-generation data assimilation and forecasting for weather and climate"
Starts November 2026; four years, renewable for a further three.

2026–30 **Natural Environment Research Council (NERC)** (£2.35m fEC total project value, £1.2m fEC to Reading)
Co-lead, "AUSPICE – Advancing Understanding of the Signal-to-noise Paradox and its Impacts on Climate Ensembles"
With Antje Weisheimer (lead, Oxford & ECMWF) and co-leads at Reading and Oxford.

2025 **NERC** (£50k)
Co-lead, "Data Assimilation Training Course"
With Alison Fowler (lead) and co-leads at Reading.

Publications

Asterisks (*) indicate papers on which I am the sole or co-first author, or the sole or co-corresponding author.

Books/chapters

*28. **Bach, E.**, R. Baptista, D. Sanz-Alonso, and A. Stuart (2027). *Machine Learning for Inverse Problems and Data Assimilation*. Cambridge University Press. Forthcoming (preprint on arXiv).

- Pre-publication versions used in the syllabi of courses at the University of Pennsylvania, University of Chicago, University of Edinburgh, TU Eindhoven, Utrecht University, New York University, Caltech, and University of Reading.

*27. **Bach, E.** (2027). "The Interface of Data Assimilation and Machine Learning". In: *Weather and Climate: Applications of Machine Learning and Artificial Intelligence*. Ed. by S. Driscoll, K. M. R. Hunt, L. Mansfield, R. Swaminathan, H. Wei, **E. Bach**, and A. Peard. Elsevier. Forthcoming (preprint on arXiv).

In review/preprints

*26. **Bach, E.** (2026). "Analysis Increments for Model Bias Diagnosis and Correction: Theoretical Foundations". arXiv (under review in *Quarterly Journal of the Royal Meteorological Society*).

*25. **Bach, E.**, R. Baptista, J. Bröcker, B. Chen, and A. Stuart (2026 preprint). "Learning Probabilistic Filters with Strictly Proper Scoring Rules". *Journal of Machine Learning Research* (accepted with minor revisions).

*24. **Bach, E.**, R. Baptista, E. Luk, and A. Stuart (2025 preprint). "Learning Optimal Filters Using Variational Inference". arXiv.

- Spotlight talk at Machine Learning for Earth System Modeling (ML4ESM) Workshop at ICML 2024.

Published (see Google Scholar)

*23. Bröcker, J. and **E. Bach** (2026). "A Generalisation of the Signal-to-Noise Ratio Using Proper Scoring Rules". *Meteorological Applications*.

22. Arcucci, R. et al. (2026). "The Convergence of Machine Learning and Data Assimilation in Earth System Science". *npj Artificial Intelligence*.

21. Li, Z. et al. (2026). "National Assessment Reveals Widespread Wind Farm Impacts on Land Surface Temperature and Vegetation in China". *Geography and Sustainability*.

20. Deck, K. et al. (2026). "ClimaLand: A Land Surface Model Designed to Enable Data-Driven Parameterizations". *Journal of Advances in Modeling Earth Systems*.

19. **Bach, E.**, R. Baptista, E. Calvello, B. Chen, and A. Stuart (2026). "Learning Enhanced Ensemble Filters". *Journal of Computational Physics*.

*18. **Bach, E.**, D. Crisan, and M. Ghil (2025). "Forecast Error Growth: A Dynamic–Stochastic Model". *Chaos: An Interdisciplinary Journal of Nonlinear Science*.

*17. Vernon, S., **E. Bach**, and O. R. A. Dunbar (2025). "Nesterov Acceleration for Ensemble Kalman Inversion and Variants". *Journal of Computational Physics*.

16. Vishny, D., M. Morzfeld, K. Gwirtz, **E. Bach**, O. R. A. Dunbar, and D. Hodyss (2024). "High-Dimensional Covariance Estimation From a Small Number of Samples". *Journal of Advances in Modeling Earth Systems*.

    - Top 10% most-viewed papers published in 2024.

*15. Manta, G., **E. Bach**, S. Talento, M. Barreiro, S. Speich, and M. Ghil (2024). "The South Atlantic Dipole via Multichannel Singular Spectrum Analysis". *Scientific Reports*.

*14. **Bach, E.**, V. Krishnamurthy, S. Mote, J. Shukla, A. S. Sharma, E. Kalnay, and M. Ghil (2024). "Improved Subseasonal Prediction of South Asian Monsoon Rainfall Using Data-Driven Forecasts of Oscillatory Modes". *Proceedings of the National Academy of Sciences*.

    - Featured in the UN WIPO *Green Technology Book*.
    - Written about by UN Office for Disaster Risk Reduction.

*13. **Bach, E.**, T. Colonius, I. Scherl, and A. Stuart (2024). "Filtering Dynamical Systems Using Observations of Statistics". *Chaos: An Interdisciplinary Journal of Nonlinear Science*.

    - *Chaos* Editor's Pick.

12. Xu, Z., Y. Li, Y. Qin, and **E. Bach** (2024). "A Global Assessment of the Effects of Solar Farms on Albedo, Vegetation, and Land Surface Temperature Using Remote Sensing". *Solar Energy*.

    - 2025 *Solar Energy* Best Paper Award in the Solar Energy Systems Integration subject area.
    - Written about by *PV Magazine* and *Climate Feedback*.

*11. **Bach, E.** and M. Ghil (2023). "A Multi-Model Ensemble Kalman Filter for Data Assimilation and Forecasting". *Journal of Advances in Modeling Earth Systems*.

    - In top 10% most-viewed papers published by the journal in 2023.

10. Chattopadhyay, A., E. Nabizadeh, **E. Bach**, and P. Hassanzadeh (2023). "Deep Learning-Enhanced Ensemble-Based Data Assimilation for High-Dimensional Nonlinear Dynamical Systems". *Journal of Computational Physics*.

    - Featured in *SIAM News Blog*.

9. Dunbar, O. R. A., I. Lopez-Gomez, A. Garbuno-Iñigo, D. Z. Huang, **E. Bach**, and J.-l. Wu (2022). "EnsembleKalmanProcesses.Jl: Derivative-free Ensemble-Based Model Calibration". *Journal of Open Source Software*.

8. Chattopadhyay, A., M. Mustafa, P. Hassanzadeh, **E. Bach**, and K. Kashinath (2022). "Towards Physics-Inspired Data-Driven Weather Forecasting: Integrating Data Assimilation with a Deep Spatial-Transformer-Based U-NET in a Case Study with ERA5". *Geoscientific Model Development*.

7. Qin, Y., Y. Li, R. Xu, C. Hou, A. Armstrong, **E. Bach**, Y. Wang, and B. Fu (2022). "Impacts of 319 Wind Farms on Surface Temperature and Vegetation in the United States". *Environmental Research Letters*.

    - Featured by *Skeptical Science* (climate change education website).

*6. **Bach, E.**, S. Mote, V. Krishnamurthy, A. S. Sharma, M. Ghil, and E. Kalnay (2021). "Ensemble Oscillation Correction (EnOC): Leveraging Oscillatory Modes to Improve Forecasts of Chaotic Systems". *Journal of Climate*.

*5. **Bach, E.** (2021). "parasweep: A Template-Based Utility for Generating, Dispatching, and Post-Processing of Parameter Sweeps". *SoftwareX*.

*4. **Bach, E.**, S. Motesharrei, E. Kalnay, and A. Ruiz-Barradas (2019). "Local Atmosphere–Ocean Predictability: Dynamical Origins, Lead Times, and Seasonality". *Journal of Climate*.

    - 3rd most-read article in the *Journal of Climate* from 5 June 2019 – 5 June 2020.

3. Penny, S. G., **E. Bach**, K. Bhargava, C.-C. Chang, C. Da, L. Sun, and T. Yoshida (2019). "Strongly Coupled Data Assimilation in Multiscale Media: Experiments Using a Quasi-Geostrophic Coupled Model". *Journal of Advances in Modeling Earth Systems*.

2. Li, Y., E. Kalnay, S. Motesharrei, J. Rivas, F. Kucharski, D. Kirk-Davidoff, **E. Bach**, and N. Zeng (2018). "Climate Model Shows Large-Scale Wind and Solar Farms in the Sahara Increase Rain and Vegetation". *Science*.

    - 14th top Earth science article on Altmetric in 2018, 79th of all articles in 2018, most-blogged about article of September 2018.
    - 10th most featured climate paper in the media in 2018, according to *Carbon Brief*.
    - Cited in IPCC Sixth Assessment Report and IPCC Special Report on Climate Change and Land.
    - Written about by the UN Environment Programme, BBC, *Los Angeles Times*, *Popular Science*, and other international media.

*1. **Bach, E.**, V. Radić, and C. Schoof (2018). "How Sensitive Are Mountain Glaciers to Climate Change? Insights from a Block Model". *Journal of Glaciology*.

    - Shortlisted for 2020 IACS-IGS Graham Cogley Award, for papers published by early career scientists in the *Journal of Glaciology* or *Annals of Glaciology*.
    - Selected glaciology article for the period 2015–2019, Canadian National Committee for the International Union of Geodesy and Geophysics.

Honours and Awards

2026–30/33 **Future Leaders Fellowship**, UK Research and Innovation

2022–4 **Stanback Postdoctoral Fellowship in Global Environmental Science**, Division of Geological and Planetary Sciences, Caltech, Pasadena, USA

2021–2 **Make Our Planet Great Again Postdoctoral Fellow**, Campus France, République Française

2020 **Eugene Rasmusson Fellowship**, Department of Atmospheric and Oceanic Science, University of Maryland, College Park, USA

2020–1 **Ann G. Wylie Dissertation Fellowship**, University of Maryland, College Park, USA

2017–21 **Flagship Fellowship**, University of Maryland, College Park, USA

Teaching

Postgraduate

2025,6 **MTMCW: Causality and Decision-Making**, Department of Meteorology, Reading
Taught lectures and practicals on causal discovery for climate science.

2025 **Machine Learning for Inverse Problems**, Mathematics for Our Future Climate CDT (Reading, Imperial, and Southampton students)
Designed and taught short PhD course on the mathematical foundations of machine learning applied to Bayesian inverse problems.

2024 **ACM270: Inverse Problems and Data Assimilation, a Machine Learning Approach**, Department of Computing and Mathematical Sciences, Caltech
Creating curriculum and co-teaching graduate-level, special topics course with Andrew Stuart.

Undergraduate

2025,6 **ST2PST: Probability and Statistical Theory** (50%), Department of Mathematics and Statistics, Reading
Taught statistical theory portion.

2026 **MA1LA: Linear Algebra** (academic tutor), Department of Mathematics and Statistics, Reading

2025,6 **MA3PPR: Portfolio of Projects**, Department of Mathematics and Statistics, Reading
Supervised third-year projects.

Short Courses/Tutorials

2026 **Data Assimilation Summer School**, WMO Symposium on Data Assimilation & International Symposium on Data Assimilation, Nanjing, China
Taught lectures on machine learning in data assimilation.

2025 **Tutorial on Data Assimilation**, University of Warsaw, Warsaw, Poland
Designed and taught short course on machine learning applied to data assimilation to mathematicians and PhD students.

2025 **NERC/NCEO/DARC Training Course on Data Assimilation and its Interface with Machine Learning**, Reading
Taught lectures and practical on machine learning in data assimilation.

2018,9 **Snakes on a Satellite workshops at AGU Fall Meetings and NOAA**, American Geophysical Union and NOAA Center for Weather and Climate Prediction, College Park, Maryland, USA
Co-taught interactive workshops on the scientific Python ecosystem and its use for Earth science data processing and analysis. Attended by over 70 participants at 2019 AGU Fall Meeting and over 100 NOAA scientists.

Supervision and Hosting

PhD students

2026– **Charlotte Gambardella** (Reading, Mathematics for Our Future Climate [MFC] CDT, Department of Mathematics and Statistics)
Co-advised with Jeroen Wouters (primary), Laura Baker, Francesco Ragone (Leicester), Nick Dunstone (Met Office).

2025– **Maori Inagawa** (Reading, MFC CDT, Department of Mathematics and Statistics)
Primary advisor, co-advised with Jochen Bröcker, Andrew Duncan (Imperial).

2025– **Cadan Plasa** (Reading, MFC CDT, Department of Mathematics and Statistics)
Co-advised with Jochen Bröcker (primary) and Antje Weisheimer (Oxford, ECMWF).

2025– **Jesse Gilbert** (Reading, Advancing the Frontiers of Earth System Prediction DTP, Department of Meteorology)
Primary advisor, co-advised with Sarah Dance, Amos Lawless, Massimo Bonavita (ECMWF), and Chris Thomas (Met Office).

Visiting PhD students

2026–7 **Amane Kubo** (University of Tokyo), one-year visit

Visiting researchers

2026–7 **Enda Zhu** (Chinese Academy of Meteorological Sciences), one-year visit

MSc students

2027 **Harriett Du Four** (Reading, Department of Meteorology)

2025 **Yuchen Yan** (Reading, Department of Meteorology)
Co-advised with Sarah Dance (primary), Guannan Hu, Rishabh Bhatt, and Mihai Alexe (ECMWF).

Undergraduate students

2026– **Xingfei Li** (Oxford)

2026– **Alvin Chung** (Imperial)

2024–5 **Enoch Luk** (Caltech; now Machine Learning Engineer at Instagram)

2024 **Kota Okuda** (Tohoku University; now PhD student at Institut Polytechnique de Paris), Summer Undergraduate Research Fellowship (SURF), co-advised with Oliver Dunbar.

2022,4 **Anagha Satish** (Caltech; now PhD student at Harvard), SURF, co-advised with Oliver Dunbar.

2023 **Sydney Vernon** (Caltech; now postbac at the University of Washington), SURF, co-advised with Oliver Dunbar.

2019–20 **Greta Easthom** (Maryland; now PhD student at North Carolina State University), undergraduate thesis, co-advised with Safa Mote.

Presentations

Invited talks

2027 **International Congress on Industrial and Applied Mathematics** (upcoming), The Hague, Netherlands

2027 **Institute of Mathematics and its Applications Seminar** (upcoming), Aldermaston, UK

2026 **Xiong'an International Symposium on Meteorological Artificial Intelligence** (upcoming, keynote), China Meteorological Administration and ECMWF, Xiong'an, China

2026 **Mathematics for Climate, Environment and Sustainability Meeting** (upcoming, keynote), Institute of Mathematics and its Applications, Didcot, UK

2026 **Statistics, Numerical Analysis, and Data Science Seminar**, University of Bath, Bath, UK

2026 **Algorithms and Computationally Intensive Inference Seminar**, University of Warwick, Coventry, UK

2026 **UK-Vietnam Meeting on Mathematics for the Environment**, Vietnam Institute for Advanced Study in Mathematics, Hanoi, Vietnam (online talk)

2026 **CAMS Seminar**, Chinese Academy of Meteorological Sciences, Beijing, China

2026 **Institute of Atmospheric Physics Seminar**, Chinese Academy of Sciences, Beijing, China

2026 **School of Geographical Science Seminar**, Beijing Normal University, Beijing, China

2026 **Engineering Seminar**, Westlake University, Hangzhou, China

2026 **American Institute of Mathematical Sciences Conference**, Athens, Greece

2026 **Extremes and Rare Events in Climate and Related Applications**, Reading, UK

2026 **Atmospheric Physics Seminar**, Imperial College London, London, UK

2026 **Mathematics Colloquium**, University of British Columbia, Vancouver, Canada

2026 **Mathematics and Climate Research Network Colloquium**, American Institute of Mathematics (online)

2025 **Land Data Assimilation Community Virtual Workshop**, Analysis, Integration, and Modeling of the Earth System (AIMES), Future Earth (online)

2025 **Probabilistic AI for Environmental Science**, Isaac Newton Institute for Mathematical Sciences, Cambridge, UK

2025 **Surrogates and Dimension Reduction in Scientific Machine Learning**, Centre for AI Fundamentals, University of Manchester, Manchester, UK (online talk)

2025 **Applied Inverse Problems**, Rio de Janeiro, Brazil

2025 **Workshop on Uncertainty Quantification for Dynamical Modelling**, School of Mathematics, University of Edinburgh, Edinburgh, UK

2025 **Accelerating statistical inference and experimental design with machine learning**, Isaac Newton Institute for Mathematical Sciences, Cambridge, UK

2024 **Computational and Methodological Statistics (CMStatistics)**, London, UK (online talk)

2024 **Dynamics Days Europe**, Bremen, Germany

2024 **AMS-UMI International Joint Meeting**, Unione Matematica Italiana and American Mathematical Society, Palermo, Italy

2024 **Chaos, Computation, Analysis and Optimization (CaCAO) Days**, Scripps Institution of Oceanography, La Jolla, USA

2024 **Applied Mathematics Seminar**, University of California, Santa Cruz, USA

2024 **SIAM Uncertainty Quantification**, Society for Industrial and Applied Mathematics, Trieste, Italy

2023 **International Congress on Industrial and Applied Mathematics**, Tokyo, Japan

2023 **SIAM Geosciences**, Society for Industrial and Applied Mathematics, Bergen, Norway

2023 **Energy and Machine Learning Seminar**, Pacific Northwest National Laboratory, Richland, USA (online)

2023 **Atmosphere Ocean Science Colloquium**, Courant Institute of Mathematical Sciences, New York University, New York City, USA

2023 **Scientific Computing Seminar**, Florida State University, Tallahassee, USA

2022 **Applied Math Seminar**, Florida International University, Miami, USA (online)

2022 **Applied Math Seminar**, Hunter College, City University of New York, New York City, USA (online)

2022 **Forecast Verification and Data Assimilation in Intermediate and Large Scale Models of Geophysical Fluid Dynamics**, Isaac Newton Institute for Mathematical Sciences, Reading, UK

2020,2 **Atmospheric, Oceanic & Earth Sciences Seminar**, George Mason University, Fairfax, USA (online)

2022 **AI for Climate Seminar**, Sorbonne University, Paris, France

Contributed talks

2027 **Workshop on Systematic Errors in Weather and Earth System Models** (upcoming), World Climate Research Programme, Pune, India (online talk)

2026 **WMO Symposium on Data Assimilation & International Symposium on Data Assimilation**, Nanjing, China

2026 **Optimal Transport in Atmosphere and Ocean Problems**, Centre for Mathematics of Planet Earth, University of Reading, Reading, UK

2025 **International Workshop on Monsoons**, World Meteorological Organization, Pune, India (online talk)

2024 **International Symposium on Data Assimilation**, Kobe, Japan

2024 **Oxford Workshop on Model Uncertainty**, Oxford University, Oxford, UK

2024 **Machine Learning for Earth System Modeling Workshop**, International Conference on Machine Learning (ICML), Vienna, Austria (online talk, joint with Enoch Luk)

2024 **RMetS Annual Weather and Climate Conference**, Royal Meteorological Society, Reading, UK

2024 **Toward Minimizing Early Model Biases and Errors in S2S Predictions**, National Oceanic and Atmospheric Administration, Boulder, USA (online talk)

2024 **Conference on Hurricanes and Tropical Meteorology**, American Meteorological Society, Long Beach, USA

2024 **Workshop on Machine Learning for Earth System Observation and Prediction**, European Space Agency and ECMWF, Frascati, Italy (online talk)

2024 **International Symposium on Data Assimilation – Predictability**, online

2024 **International Symposium on Data Assimilation – Advancements in Ensemble Data Assimilation**, online

2023 **AGU Fall Meeting**, American Geophysical Union, San Francisco, USA

2023 **EGU General Assembly**, European Geosciences Union, Vienna, Austria (online talk)

2022 **Model Hierarchies Workshop**, Stanford University, Stanford, USA

2022 **SIAM Mathematics of Planet Earth**, Society for Industrial and Applied Mathematics, Pittsburgh, USA (online talk)

2022 **International Symposium on Data Assimilation**, Colorado State University, Fort Collins, USA

2022 **EGU General Assembly**, European Geosciences Union, Vienna, Austria (online talk)

2020 **NOAA Workshop on Leveraging AI in Environmental Sciences**, online

2019 **AGU Fall Meeting**, American Geophysical Union, San Francisco, USA

Invited Meetings

2026 **Workshop on How to Spend $15 Billion on Climate Change Research**
London School of Economics, London, UK

Convening on climate research prioritisation.

2025 **Track 1.5 Dialogue on UK–India AI Opportunities**
UK Foreign, Commonwealth & Development Office and British High Commission in India, Alan Turing Institute, London, UK

Government-level dialogue on AI policy and international research collaboration.

Editorial Roles

2023–6 **Special Issue Editor, *Earth System Dynamics***
Co-editor of special issue "Theoretical and computational aspects of ensemble design, implementation, and interpretation in climate science".
Joint issue between *Earth System Dynamics*, *Nonlinear Processes in Geophysics*, and *Geoscientific Model Development*.

Sessions Co-Chaired

2026 **EGU General Assembly 2026**, Mathematics of Planet Earth session, European Geosciences Union, Vienna, Austria

2024 **SIAM Uncertainty Quantification**, Advances at the intersection of probabilistic machine learning, inverse problems and data assimilation mini-symposium, Society for Industrial and Applied Mathematics, Trieste, Italy

2023 **SIAM Geosciences**, Theoretical Aspects of Ensemble Design and Interpretation in Climate Science and Modelling session, Society for Industrial and Applied Mathematics, Bergen, Norway

2022 **2022 Model Hierarchies Workshop**, Model Hierarchies for the Ocean 2 session, Stanford, USA

2022 **SIAM Mathematics of Planet Earth**, Data Science session, Pittsburgh, USA

Partnerships

2026–9 **Advancing the Frontiers of Earth System Prediction Fellowship** (Mark Muetzelfeldt, Reading)
Project partner, "ALOFT: ALigning Organized Features between coarse- and high-resolution simulaTions"

Thesis Examination

2025 **Reyk Börner** (Reading, Department of Mathematics and Statistics), internal examiner

Visiting Positions

2025 **Residential Programme participant, Isaac Newton Institute Programme on Representing, Calibrating & Leveraging Prediction Uncertainty from Statistics to Machine Learning** (June 19 – July 8), Cambridge, UK

2022 **Residential Programme participant, Isaac Newton Institute Satellite Programme on Geophysical Fluid Dynamics** (September 10–25), Reading, UK

2018 **Visiting Researcher, Indian Institute of Tropical Meteorology** (November 18 – December 8), Pune, India

Peer Review

- Over 60 journal reviews, including *Science*; *Proceedings of the National Academy of Sciences*; *Nature Machine Intelligence*; *Chaos*; *Journal of Computational Physics*; *Physics of Fluids*; *Journal of Climate*; *Journal of the Atmospheric Sciences*; *Geophysical Research Letters*; *Journal of Advances in Modeling Earth Systems*; *Bulletin of the American Meteorological Society*; *Quarterly Journal of the Royal Meteorological Society*; *Geoscientific Model Development*; *Monthly Weather Review*; *Climate Dynamics*; and *Nonlinear Processes in Geophysics*.
- Grant reviewer for National Research, Development and Innovation Office, Hungary (2022).

Service and Engagement

Policy Impact

My publications have been cited in 23 policy documents, including the IPCC Sixth Assessment Report and the IPCC Special Report on Climate Change and Land.

EDI/Outreach Roles

2026– **LGBTQ+ Working Group Member, Wellbeing, Inclusion, Diversity and Equality (WIDE) Committee**, School of Mathematical, Physical and Computational Sciences, University of Reading, Reading, UK

2024– **Outreach Officer, Centre for Doctoral Training in Mathematics for Our Future Climate**, University of Reading, Reading, UK
Organising outreach events, including mathematical outreach for school students with Axiom Maths.

2019–20 **President of Science for the People**, University of Maryland, College Park, USA
Organised events on issues of social justice in science for students and other members of the public, including on climate change and on implicit bias in artificial intelligence.

Blog and magazine articles

- **Bach, E.** and O. Dunbar (2023). "How Do We Estimate Climate Parameters? An Introduction to Ensemble Kalman Inversion". *Climate Modeling Alliance Blog*.
- Chattopadhyay, A., E. Nabizadeh, **E. Bach**, and P. Hassanzadeh (2021). "Deep Learning-Augmented Data Assimilation for Next-Generation Predictive Models". *SIAM News Blog*.
- Pentakota, S., S. V. Gade, Suryachandra A. Rao, C. Da, K. Bhargava, C.-C. Chang, **E. Bach**, E. Kalnay, and T. Sluka (2020). "Advances in Coupled Data Assimilation, Ensemble Forecasting, and Assimilation of Altimeter Observations". *CLIVAR Exchanges*.

Software

- **EnsembleKalmanProcesses.jl** (contributor and reviewer)
  Implemented new methods, including localisation, in CliMA's library for derivative-free model calibration, used by research groups at MIT, Stanford, and Scripps Institution of Oceanography.
- **SageMath** (contributor)
  Code merged into several releases of the open-source mathematics system, focusing on improvements in symbolic computation and special functions.
- **parasweep** (author)
  Utility for generating, dispatching and post-processing parameter sweeps.